Did Medicaid Expansion Release Job Lock?

Coverage, Job Mobility, and Labor Supply under Staggered Adoption

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Abstract

I estimate the effects of the Affordable Care Act Medicaid expansions on health coverage and on outcomes that the "job lock" hypothesis predicts should respond when coverage no longer depends on employment: holding multiple jobs, self-employment, voluntary part-time work, employment, and hours. Using the CPS Annual Social and Economic Supplement for reference years 2008–2024, a state-year panel of adults aged 26–64 with family income at or below 138 percent of the federal poverty level in 40 states, and the Callaway and Sant'Anna (2021) estimator for staggered adoption, I find that expansion raised Medicaid coverage among states that expanded in 2014 by 13.3 percentage points (s.e. 1.5) and reduced uninsurance by 9.9 points (s.e. 1.4), averaged over the first five years. These coverage effects are stable across comparison periods, control groups, age ranges, timing definitions, and sample windows. In contrast, I find no robust evidence that expansion changed job mobility, self-employment, part-time work, employment, or hours. In the preferred specification the employment effect is -0.5 points, with a 95 percent confidence interval of -1.9 to 1.0; this rules out employment declines larger than about 0.15 per new Medicaid enrollee. A methodological finding is that the survey year immediately before expansion appears to be partly treated: the CPS-ASEC is fielded in February–April, after January expansions take effect, and its redesigned questionnaire uses current coverage to help respondents recall the previous year. Measuring effects against that year, the convention in standard event-study software, understates the coverage gain by about a quarter (Medicaid) and more than half (uninsurance) and produces apparent positive effects on employment and voluntary part-time work that disappear once the comparison year is moved back. I discuss implications for the Medicaid work requirements that states must implement by January 2027.

Keywords: Medicaid expansion; job lock; labor supply; staggered difference-in-differences; event study; CPS-ASEC.
JEL codes: I13, I18, J22, J62, H75.

1. Introduction

In the United States most working-age adults obtain health insurance through an employer. When coverage is tied to a job, workers may stay in positions they would otherwise leave, avoid starting a business, or keep working full time to keep their benefits. This distortion, "job lock," has been studied for three decades (Madrian, 1994; Gruber and Madrian, 2002). A public coverage option that does not depend on employment should loosen it. The Affordable Care Act (ACA) Medicaid expansion, which extended eligibility to adults with family income up to 138 percent of the federal poverty level (FPL) in states that chose to adopt it, offers a large policy experiment for testing that prediction among low-income workers.

The same expansion is also at the center of current policy. Public Law 119-21, enacted in July 2025, requires adults covered through the expansion group to document 80 hours a month of work or qualifying activity. Under the interim final rule issued by the Centers for Medicare & Medicaid Services (CMS) on June 1, 2026, states must generally implement the requirement no later than January 1, 2027 (Centers for Medicare & Medicaid Services, 2026). The Congressional Budget Office projects that the requirement alone will increase the number of uninsured people by about 5.3 million in 2034 (Center for Children and Families, 2025). Whether expansion reduced work in the first place, and whether it gave low-income workers more freedom to move between jobs, are therefore questions with direct budgetary and coverage consequences.

This paper asks two questions. First, how much did expansion change health coverage for low-income adults? Second, did it change the labor-market margins that job-lock theory says should respond: the number of employers a person works for in a year, self-employment, part-time work (separately for voluntary and involuntary reasons), whether the person worked at all, and annual hours? I use the CPS Annual Social and Economic Supplement (CPS-ASEC) from IPUMS (Flood et al., 2025) for reference years 2008–2024, build a balanced state-year panel of adults aged 26–64 with family income at or below 138 percent of FPL in 40 states, and estimate effects with the Callaway and Sant'Anna (2021) estimator. That estimator compares each adoption cohort only with states that have not yet been treated. It avoids the "forbidden comparisons" that make two-way fixed effects (TWFE) regressions unreliable when adoption is staggered and effects vary across cohorts or over time (Goodman-Bacon, 2021; de Chaisemartin and D'Haultfœuille, 2020; Sun and Abraham, 2021).

The paper makes three contributions. First, it provides a transparent, fully replicable estimate of the coverage and labor-market effects of expansion that covers all adoption cohorts through 2024, including states that expanded after 2018. Second, it documents a measurement problem that matters for any event study built on the CPS-ASEC. The survey is fielded in February through April and asks about the previous calendar year, and since 2014 its redesigned questionnaire asks about current coverage to help respondents recall the past year (U.S. Census Bureau, 2014; Pascale et al., 2016). For a state that expanded on January 1, 2014, coverage "in 2013" is reported by people who in many cases already had Medicaid at the time of interview. The year before expansion is thus partly treated. The usual event-study convention, which measures every effect against the year before treatment, uses exactly this contaminated year as the reference. I show that doing so understates the coverage gain by a quarter to more than a half and creates spurious labor-market "effects." Moving the comparison back one year removes both problems. Using the average of the pre-period removes the coverage bias but leaves a modest positive employment estimate, which shows how sensitive labor-market estimates are to the choice of baseline years. Third, I subject the labor-market estimates to a battery of checks: alternative comparison years, control groups, cohorts, samples, sample

windows, TWFE and its decomposition, and two placebo designs. The paper reports which results survive and which do not.

The main findings are as follows. Expansion produced large, robust coverage gains. In the preferred specification (states that expanded in 2014, compared with never-expanding states, with effects measured against two years before expansion), Medicaid coverage rose by 13.3 percentage points and uninsurance fell by 9.9 points over the first five years. Coverage through a private plan held in the person's own name fell by 3.3 points, consistent with some substitution away from private coverage. None of the job-lock outcomes moves in the direction the theory predicts in a way that survives the robustness checks. The employment estimate is close to zero and precise enough to rule out declines larger than about 1.9 percentage points; measured against the pre-period average it is a modest +2.0 points, so a small increase cannot be excluded. In the preferred specification, the estimates for voluntary and involuntary part-time work and for multiple employers are statistically significant, but they point away from job-lock release. Similar shifts are already visible in the year before expansion, and the all-cohort versions disappear when the sample ends in 2022. I therefore do not interpret them as effects of expansion.

These results agree with the large body of evidence that the ACA expansions did not reduce employment (Kaestner et al., 2017; Leung and Mas, 2018; Gooptu et al., 2016; Duggan et al., 2019; Buchmueller et al., 2021) and with the experimental finding that Medicaid coverage did not reduce employment in Oregon (Baicker et al., 2014). They contrast with the strong employment responses to the loss of public coverage in Tennessee documented by Garthwaite et al. (2014), which concerned a different population, a different policy direction, and a different period. They also complement Aslim (2022), who finds more voluntary moves from full-time to part-time work at the 138 percent eligibility threshold using a design that compares people just above and below the cutoff; my state-level design does not detect such a shift in the population as a whole.

The rest of the paper is organized as follows. Section 2 describes the policy and the predictions of job-lock theory. Section 3 describes the data. Section 4 sets out the empirical strategy, including the comparison-year problem. Section 5 presents the main results, and Section 6 the robustness checks. Section 7 discusses limitations, Section 8 policy implications, and Section 9 concludes.

2. Policy Background and Conceptual Framework

2.1 The ACA Medicaid expansion

The ACA extended Medicaid eligibility to nearly all non-elderly adults with family income at or below 138 percent of FPL. The Supreme Court's 2012 decision in *NFIB v. Sebelius* made the expansion optional for states, which produced staggered adoption. Most adopting states expanded on January 1, 2014. Others followed between 2015 and 2016, and a second wave adopted between 2019 and 2023, several through ballot initiatives. Ten states in the analysis sample had not expanded by the end of 2024: Alabama, Florida, Georgia, Kansas, Mississippi, South Carolina, Tennessee, Texas, Wisconsin, and Wyoming. Table 1 lists the cohorts used in this paper.

Two features of the policy environment matter for interpretation. First, the ACA marketplaces opened in every state in 2014. In non-expansion states, adults with income between 100 and 138 percent of FPL became eligible for subsidized marketplace plans, while in expansion states the same people became eligible

for Medicaid. The estimates therefore measure the effect of Medicaid eligibility relative to a counterfactual that includes subsidized private coverage for part of the target population, not relative to no reform. Second, the period after 2020 includes the pandemic, the continuous Medicaid enrollment provision of 2020–2023, and the subsequent "unwinding." I test sensitivity to ending the sample before these periods.

2.2 What job-lock theory predicts

If workers value health insurance and can obtain it only through an employer, coverage acts as a tax on leaving a job with benefits. A public option that does not depend on employment removes that tax. For low-income adults the theory yields several predictions:

- **Job mobility** should rise: workers are freer to leave a job, so more people should work for more than one employer during a year.
- **Self-employment** should rise, since starting a business no longer means giving up coverage.
- **Voluntary part-time work** should rise among workers who were working full time mainly to keep benefits.
- **Employment and hours** have ambiguous predictions. Eligibility capped at 138 percent of FPL creates an incentive to keep income low, and removing the need for employer coverage removes one reason to work. Better health and financial security can raise work capacity.

A caveat is important for this population. Relatively few adults below 138 percent of FPL hold employer coverage in their own name, so the scope for classic job lock is smaller than among middle-income workers. Expansion may matter more for "employment lock" (working to obtain coverage at all) than for mobility between jobs (Garthwaite et al., 2014).

3. Data

3.1 Sample

I use the CPS-ASEC samples from survey years 2009 through 2025, extracted from IPUMS CPS (Flood et al., 2025). Each survey year t describes coverage and work in calendar year $t-1$, the *reference year*, so the data span reference years 2008–2024. I keep ASEC records with positive supplement weights. For survey year 2014, which was fielded on two overlapping samples, I keep the 5/8 file (HFLAG $= 0$), which Census and IPUMS recommend for comparisons with earlier years (Flood et al., 2025). A direct comparison shows the choice does not matter for Medicaid: among adults aged 26–64 the weighted Medicaid rate is 11.02 percent in the 5/8 file and 11.26 percent in the 3/8 file.

The main sample consists of adults aged 26–64, which excludes adults aged 19–25 because many gained dependent coverage in 2010. I exclude members of the Armed Forces and keep people whose family income is at or below 138 percent of FPL. Because the IPUMS poverty variable is grouped into intervals, I compute the income-to-poverty ratio directly from family income and family size using the Department of Health and Human Services poverty guidelines for each year. I exclude seven jurisdictions that expanded early under ACA options or county programs (California, Colorado, Connecticut, the District of Columbia, Minnesota, New Jersey, Washington) and four states that already covered low-income adults broadly before

Table 1. Expansion cohorts in the analysis sample

Cohort (first treated year)	States	State list
2014	15	AR, AZ, HI, IA, IL, KY, MD, MI, ND, NM, NV, OH, OR, RI, WV
2015	3	IN, NH, PA
2016	3	AK, LA, MT
2019	2	ME, VA
2020	2	ID, UT
2021	2	NE, OK
2022	1	MO
2023	1	SD
2024	1	NC
Never expanded (through 2024)	10	AL, FL, GA, KS, MS, SC, TN, TX, WI, WY

Notes: A state is assigned to the first reference year in which expansion was in effect for at least six months. Michigan (April 2014) is in the 2014 cohort; New Hampshire (August 2014) is in the 2015 cohort; Alaska (September 2015) and Louisiana (July 2016) are in the 2016 cohort; Nebraska (October 2020) is in the 2021 cohort; Missouri (eligibility retroactive to July 2021, enrollment from October 2021) is in the 2022 cohort; North Carolina (December 2023) is in the 2024 cohort. Excluded: early expanders CA, CO, CT, DC, MN, NJ, WA and prior-coverage states DE, MA, NY, VT.

2014 (Delaware, Massachusetts, New York, Vermont). The final sample contains 191,112 person-records representing 333.2 million weighted person-years in 40 states, which I collapse to 680 state-year cells (40 states $\times$ 17 years) using the ASEC weights.

3.2 Treatment timing

A state's cohort g is the first reference year in which expansion was in effect for at least half the year: an expansion effective on or before July 1 is assigned to that year and a later one to the following year. Table 1 shows the result. The 2014 cohort contains 15 states. Every other cohort contains between one and three states. An alternative coding that assigns each state to the year in which expansion took effect is used as a robustness check.

3.3 Outcomes

All outcomes are defined for every person in the sample, with non-workers coded as zero for work-related outcomes. This avoids conditioning on employment, which expansion could itself change.

- **Coverage:** Medicaid coverage during the year; uninsured (no coverage of any kind); a private plan held in the person's own name; and an employer plan held in the person's own name. The last measure comes from a question that IPUMS reports only for reference years 2008–2017, so it is analyzed in that window only.
- **Job mobility and job lock:** working for two or more employers during the year (the number of employers is top-coded at three in the CPS); self-employment in the longest job held; and part-time work for voluntary (non-economic) and involuntary (could not find full-time work, slack work) reasons.
- **Labor supply:** worked at all during the year, and annual hours (weeks worked times usual weekly hours).

The state-year panel is weighted by each state's average pre-2014 low-income population, normalized to a mean of one. The weights are fixed over time because post-expansion populations are themselves affected

by the policy: expansion can change who has income below the threshold.

Figure 1 plots the raw means of six outcomes for the 2014 cohort and the never-expanding states. Before 2014, Medicaid coverage was about 8 to 10 points higher, and uninsurance about 10 points lower, in states that would expand. Both groups show a visible break in reference year 2013, before the first expansions, which anticipates the measurement problem discussed below.

4. Empirical Strategy

4.1 Group-time effects and aggregation

Let Y_{st} be an outcome for state s in reference year t, and let G_s be the cohort, with $G_s = 0$ for never-treated states. Following Callaway and Sant'Anna (2021), the building block is the average effect for cohort g in year t,

$$ATT(g,t) = \mathbb{E}\left[Y_t - Y_{b(g)} \mid G = g\right] - \mathbb{E}\left[Y_t - Y_{b(g)} \mid G = 0\right], \tag{1}$$

where $b(g)$ is a fixed pre-treatment comparison year for cohort g and expectations are weighted by the fixed state weights. With no covariates, the doubly robust estimator reduces to this weighted difference-in-differences. Identification requires that, absent expansion, outcomes in expansion and never-expansion states would have followed parallel paths from year $b(g)$ onward, and that outcomes are not affected before $b(g)$.

Event-study coefficients average the group-time effects across cohorts by years since expansion, $e = t - g$, weighting cohorts by size:

$$\theta(e) = \sum_g w_g(e)\, ATT(g, g+e). \tag{2}$$

The headline statistic in every table is the average effect over the first five years of exposure, $\bar{\theta} = \frac{1}{5}\sum_{e=0}^{4} \theta(e)$. Fixing the horizon keeps the specifications comparable, since later event times rest on fewer and earlier cohorts. Standard errors come from the estimator's influence function and are clustered by state.

4.2 Why not two-way fixed effects

The conventional alternative regresses Y_{st} on a post-expansion indicator with state and year fixed effects. With staggered adoption, the TWFE coefficient is a weighted average of all two-by-two comparisons in the data, including comparisons in which states that expanded earlier serve as controls for states that expanded later (Goodman-Bacon, 2021). If the effect of expansion grows over time, as coverage effects do while enrollment builds, these comparisons subtract part of the early cohort's growing effect from the later cohort's estimate, and some cohort-years can even receive negative weight (de Chaisemartin and D'Haultfœuille, 2020). Section 6 reports TWFE estimates and decomposes them to show how much these comparisons matter here.

4.3 The comparison-year problem

Standard event-study practice sets $b(g) = g-1$, the last year before treatment. In the CPS-ASEC this choice is problematic. Survey year g is fielded in February through April of calendar year g and asks about calendar year $g-1$. Since 2014 the redesigned questionnaire asks about current coverage and uses it to help respon-

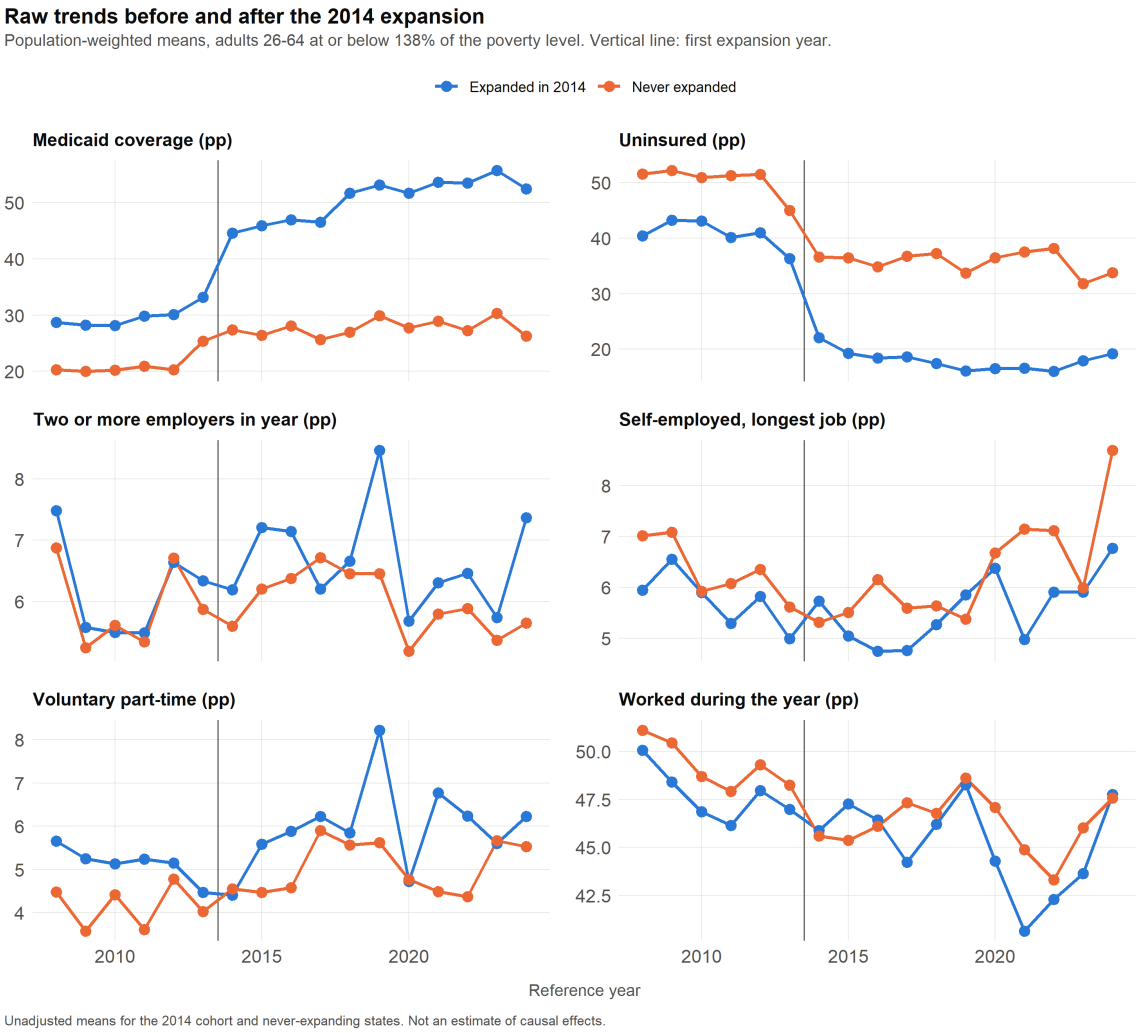


Figure 1. Raw trends, 2014 cohort and never-expanding states

Notes: Population-weighted means for adults aged 26–64 with family income at or below 138 percent of FPL. Reference years on the horizontal axis; the vertical line marks the first expansion year. Unadjusted means; not estimates of causal effects.

dents recall coverage in the previous year (U.S. Census Bureau, 2014; Pascale et al., 2016). For a state that expanded on January 1 of year g, many respondents interviewed in early year g already have Medicaid, and some of them report it as coverage in year $g-1$. The comparison year is then partly treated.

This contamination has a recognizable signature. Every event-study coefficient is measured relative to year $g-1$, so if that year is shifted, all pre-treatment coefficients are shifted by the same amount in the opposite direction, and the post-treatment effects are understated by the same amount. Section 5 shows exactly this pattern. I therefore use two alternatives:

1. **Comparison year $g-2$.** Implemented with one year of "anticipation" in the Callaway and Sant'Anna (2021) framework, which moves $b(g)$ back to $g-2$ and estimates the year-before effect $\theta(-1)$ as a free coefficient. That coefficient becomes a direct measure of the contamination.
2. **Pre-period average.** Re-reference every coefficient to the mean of the pre-treatment coefficients ($e \leq -2$), using the influence function to propagate uncertainty. This uses several baseline years and is less sensitive to any single one.

4.4 Preferred specification

The preferred specification estimates effects for the 2014 cohort (15 states) against the never-treated states (10 states), with year $g-2$ (2012) as the comparison year. I prefer it for three reasons. The 2014 cohort is the only large cohort, so its estimates do not rest on one or two states. It passes a placebo test in its own pre-period (Section 6), while a placebo built from the later-expanding states fails badly. And its comparison year, 2012, precedes both the contaminated year and the 2014 survey redesign. I report the all-cohort estimates alongside it throughout.

5. Results

5.1 The baseline event study and its warning sign

Figure 2 shows the event study under the conventional comparison year $g-1$, for all cohorts. The Medicaid panel has a clear jump at expansion, but every pre-treatment coefficient lies below zero (mean -2.7 points), and every uninsurance pre-coefficient lies above it (mean $+3.7$ points). The pre-period is not trending. It is *offset*: the reference year itself is unusual. Joint Wald tests reject the null of zero pre-treatment coefficients for eight of the ten outcomes (Appendix Table A1; seven of ten with the small-sample F correction in Table 2). This is the signature of a contaminated comparison year described in Section 4.

When year $g-2$ is the comparison, the year-before coefficient becomes an estimate (Table 2, last column). In the year before expansion, Medicaid coverage in expansion states was already 2.8 points higher (s.e. 1.5) and uninsurance 4.7 points lower (s.e. 1.8) than in non-expansion states, relative to two years before. Once that year is no longer the reference, the average pre-treatment coefficient for Medicaid moves from -2.70 to 0.07, and for uninsurance from 3.73 to -1.17.

5.2 Coverage

Table 3 reports the headline effects. In the preferred specification (column 1), Medicaid coverage rose by 13.3 percentage points (95 percent confidence interval 10.3 to 16.3) and uninsurance fell by 9.9 points (-12.6

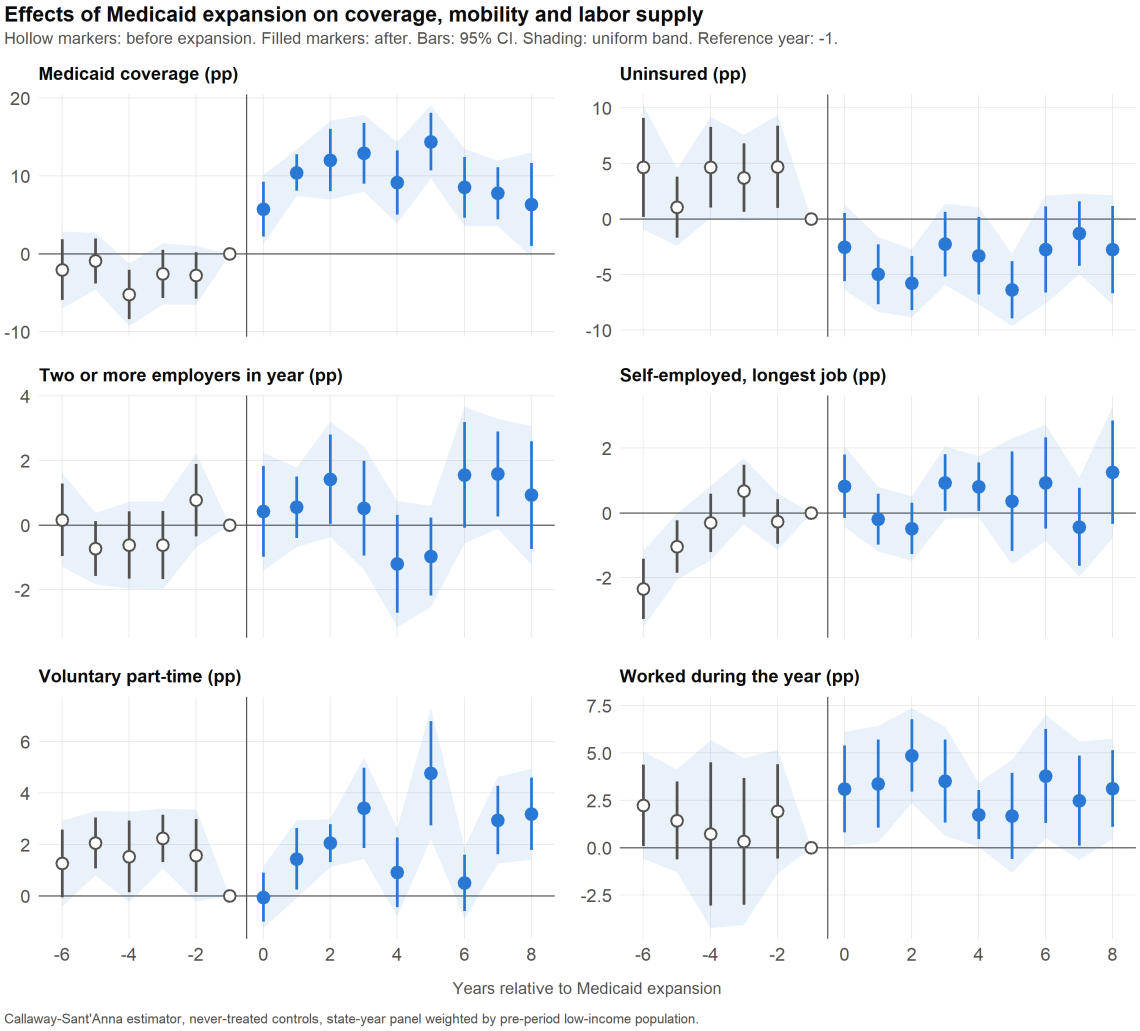


Figure 2. Event study measured against the year before expansion (baseline)

Notes: Callaway–Sant'Anna estimates, all cohorts, never-treated controls, reference year $e = -1$. Bars are pointwise 95 percent confidence intervals; shading is the uniform (sup-t) band from the multiplier bootstrap. Estimates at $e \geq 6$ rest on the 2014–2016 cohorts only.

Table 2. Pre-treatment diagnostics under the two comparison years

	Compared with year $g-1$			Compared with year $g-2$			
Outcome	Mean pre	Joint p	Slope p	Mean pre	Joint p	Slope p	Year before, $\theta(-1)$
Medicaid coverage	−2.70	0.001	0.718	0.07	< 0.001	0.805	2.76* (1.48)
Uninsured	3.73	0.189	0.959	−1.17	0.132	0.426	−4.67** (1.82)
Private plan, own name	−0.86	0.018	0.470	0.78	0.028	0.750	1.48 (0.92)
Own employer plan[a]	0.02	0.006	0.002	2.46	0.002	< 0.001	1.94** (0.94)
Two or more employers	−0.20	0.013	0.002	−1.22	0.030	0.068	−0.77 (0.55)
Self-employed	−0.66	< 0.001	< 0.001	−0.48	< 0.001	< 0.001	0.27 (0.34)
Voluntary part-time	1.73	< 0.001	< 0.001	0.20	0.450	0.473	−1.57** (0.66)
Involuntary part-time	−1.86	< 0.001	< 0.001	0.28	0.065	0.255	2.09*** (0.66)
Worked during year	1.33	0.116	0.022	−0.73	0.629	0.768	−1.91 (1.18)
Annual hours (hours)	19.7	0.519	0.342	−13.5	0.388	0.103	−30.5 (27.9)

Notes: All cohorts, never-treated controls. "Mean pre" is the average pre-treatment event-study coefficient ($e \leq -2$ when comparing with $g-1$, $e \leq -3$ when comparing with $g-2$). "Joint p" tests that all pre-treatment coefficients are zero (Wald test with a small-sample F correction for 40 clusters). "Slope p" tests for a linear pre-trend through the reference year. $\theta(-1)$ is the estimated effect in the year just before expansion; standard error in parentheses. [a] Reference years 2008–2017 only. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

to −7.3). Relative to pre-expansion levels of roughly 30 percent with Medicaid and 40 percent uninsured in these states, these are large changes. The estimates are similar for all cohorts against year $g-2$ (13.3 and −8.5), against the pre-period average (12.8 and −7.5), and with not-yet-treated states as additional controls (14.2 and −9.5). The baseline specification (last column) yields 10.1 and −3.8. Measuring against the contaminated year therefore understates the Medicaid gain by about a quarter and the fall in uninsurance by more than half.

Coverage through a private plan held in the person's own name fell by 3.3 points (s.e. 1.2) in the preferred specification, with similar estimates in every column. Part of this reflects substitution of Medicaid for private coverage. Part reflects the comparison: in non-expansion states, adults between 100 and 138 percent of FPL gained access to subsidized marketplace plans, which raises private coverage in the control group. Estimates for employer coverage in the person's own name, available only through 2017, should not be interpreted. That outcome has a strong pre-existing downward trend in expansion states (slope −0.76 points per year, $p < 0.001$; Table 2). In that shorter window the 2015 and 2016 cohorts also do not contribute estimates when the comparison year is $g-2$, which is why columns 1, 2, and 4 coincide.

5.3 Job mobility, job lock, and labor supply

The lower panels of Table 3 and Figure 3 report the labor-market outcomes. Three findings stand out.

Employment and hours show no robust effect. In the preferred specification the share who worked during the year changed by −0.47 points (95 percent CI −1.94 to 1.00) and annual hours by −2 hours (CI −34 to 30, or under 0.7 hours a week). The all-cohort estimate against year $g-2$ is 1.00 (CI −0.80 to 2.80). Scaled by the first stage, the preferred interval rules out employment declines larger than about 0.15 per new Medicaid enrollee and increases larger than about 0.08. Measured against the pre-period average, the employment estimate is larger, +2.0 (CI 0.5 to 3.5), with +30 annual hours. That version differs from the $g-2$ version because employment in expansion states was somewhat higher in 2012 than in earlier pre-period years; both comparisons are defensible, so the data rule out meaningful employment declines but

Table 3. Effects of Medicaid expansion: average over the first five years

Comparison Controls	(1) **2014 cohort** **year $g-2$** **never** **(preferred)**	(2) All cohorts year $g-2$ never	(3) All cohorts pre-mean never	(4) All cohorts year $g-2$ not-yet	(5) 2015+ cohorts year $g-2$ never	(6) All cohorts year $g-1$ never (baseline)
Coverage						
Medicaid coverage	13.28***	13.27***	12.75***	14.15***	13.25***	10.05***
	(1.53)	(1.29)	(1.27)	(1.46)	(2.33)	(1.48)
Uninsured	−9.91***	−8.45***	−7.52***	−9.52***	−5.42**	−3.79***
	(1.35)	(1.45)	(1.00)	(1.49)	(2.72)	(1.23)
Private plan, own name	−3.26***	−3.09***	−3.33***	−2.32*	−2.72	−4.19***
	(1.16)	(1.02)	(0.77)	(1.22)	(2.04)	(0.96)
Own employer plan[a]	−1.53*	−1.53*	−3.86***	−1.53*		−3.83***
	(0.91)	(0.91)	(1.12)	(0.91)		(1.40)
Job mobility and job lock						
Two or more employers	−0.93**	−0.66*	0.55*	−0.34	−0.15	0.35
	(0.38)	(0.34)	(0.30)	(0.48)	(0.55)	(0.51)
Self-employed	0.50	0.54	1.03***	0.26	0.75	0.37
	(0.31)	(0.36)	(0.19)	(0.53)	(0.85)	(0.31)
Voluntary part-time	−1.70***	−0.42	−0.18	0.23	2.19***	1.55***
	(0.40)	(0.57)	(0.42)	(0.54)	(0.72)	(0.36)
Involuntary part-time	2.91***	2.28***	1.73***	0.55	1.02	−0.13
	(0.57)	(0.49)	(0.40)	(0.96)	(0.98)	(0.44)
Labor supply						
Worked during year	−0.47	1.00	1.99***	0.85	4.00***	3.32***
	(0.75)	(0.92)	(0.75)	(0.88)	(1.39)	(0.62)
Annual hours (hours)	−2.1	8.8	30.2***	12.4	34.2	49.8***
	(16.3)	(16.3)	(10.8)	(19.4)	(33.3)	(18.6)

Notes: Each cell is the average of the event-study effects at $e = 0, \ldots, 4$, in percentage points (hours for annual hours). Callaway–Sant'Anna estimator; standard errors from the influence function, clustered by state, in parentheses. "Never" controls are the 10 never-expanding states; "not-yet" adds states that have not yet expanded. Column (5) is not estimable for employer coverage. [a] Reference years 2008–2017; in this window only the 2014 cohort contributes when the comparison year is $g-2$, so columns (1), (2), and (4) coincide. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

cannot rule out a small increase. The baseline specification instead suggests a significant 3.3-point *increase* in employment and a 50-hour increase in annual hours. These are artifacts of the comparison year. The year-before coefficient for employment is -1.9 (s.e. 1.2), so measuring against that year inflates every post-expansion difference.

The job-lock margins do not move as the theory predicts. Job-lock release predicts more voluntary part-time work, more job switching, and more self-employment. In the preferred specification voluntary part-time work *fell* by 1.7 points, involuntary part-time work rose by 2.9 points, and the share with two or more employers fell by 0.9 points. Self-employment rose by an insignificant 0.5 points, and it has a strong pre-existing upward trend (slope $+0.32$ points per year, $p < 0.001$). The part-time and mobility estimates change sign across columns of Table 3. In the all-cohort estimates against $g-2$, the shifts in part-time work are already present in the year before expansion (-1.6 for voluntary and $+2.1$ for involuntary). Involuntary part-time work goes from $+2.3$ to -0.1 when the sample ends in 2022 instead of 2024 (Section 6). Movements that begin before the policy and depend on the sample window cannot be attributed to it.

The baseline voluntary part-time effect is spurious. With year $g-1$ as the comparison, voluntary part-time work appears to rise by 1.6 points ($p < 0.001$), exactly the job-lock prediction. Against year $g-2$, against the pre-period average, and with not-yet-treated controls, the estimate is between -0.4 and $+0.2$. The baseline result reflects a dip in voluntary part-time work in expansion states in the year before expansion.

6. Robustness

6.1 Two-way fixed effects

Table 4 compares TWFE with Callaway–Sant'Anna. The static TWFE coefficient for Medicaid is 13.4, close to the Callaway–Sant'Anna estimate against year $g-2$ (13.3) and well above the baseline estimate (10.1). The same holds for uninsurance, employment, hours, and voluntary part-time work. The main reason is that the static TWFE regression compares the post-period with the average of all pre-treatment years, so the contaminated year receives little weight. In this application, most of the gap between the baseline event study and TWFE comes from the comparison year, not from heterogeneous-effects bias.

The textbook TWFE problems are present but modest. No treated state-year receives a negative weight. In the unweighted Goodman-Bacon decomposition, 54 percent of the weight comes from comparisons of treated states with never-treated states, 28 percent from earlier cohorts compared with not-yet-treated later cohorts, and 17.5 percent from "forbidden" comparisons that use already-treated states as controls. For Medicaid the forbidden comparisons average 8.2 points, against about 12.8 for the clean ones. Because the Medicaid effect grows over time, using early expanders as controls subtracts part of their effect. This pulls the TWFE estimate down by about 0.8 points, or 6 percent. TWFE also over-weights two of the three single-state late cohorts. North Carolina accounts for 0.4 percent of treated state-years but 2.6 percent of the TWFE weight, and Missouri for 1.2 percent of treated state-years but 3.7 percent of the weight.

6.2 Placebo tests

Table 5 and Figure 4 report two placebo designs in which no real expansion occurs.

- **P2 (2014-cohort pre-period).** The 15 states of the 2014 cohort are split into three groups with fake expansion dates in 2010, 2011, and 2012, and the sample ends in 2013. Against year $g-1$, none of the ten outcomes shows a significant placebo effect. Against year $g-2$, two do: Medicaid (-4.9) and private own-name coverage ($+3.1$). Both are opposite in sign to the real effects and smaller in magnitude, and they most likely reflect year-to-year noise in the pre-period.
- **P1 (late expanders).** The states that expanded in 2019–2023 are given fake expansion dates five years before their real ones (2014–2018), the sample ends in 2018, and real 2014–2016 expanders are dropped. North Carolina's fake date (2019) falls outside the window, so it serves as a control. This design fails. Five or six of ten outcomes show significant "effects," including an 8.3-point rise in employment and a 9.1-point fall in Medicaid. Each fake cohort contains one or two states, so these estimates combine two problems: genuinely different trends between late expanders and never-expanders, and standard errors that are unreliable with so few treated states.

P2 supports the comparison of the 2014 cohort with never-expanding states. P1 shows that estimates driven by the small later cohorts should not be trusted, which is why column (5) of Table 3 and cohort-specific

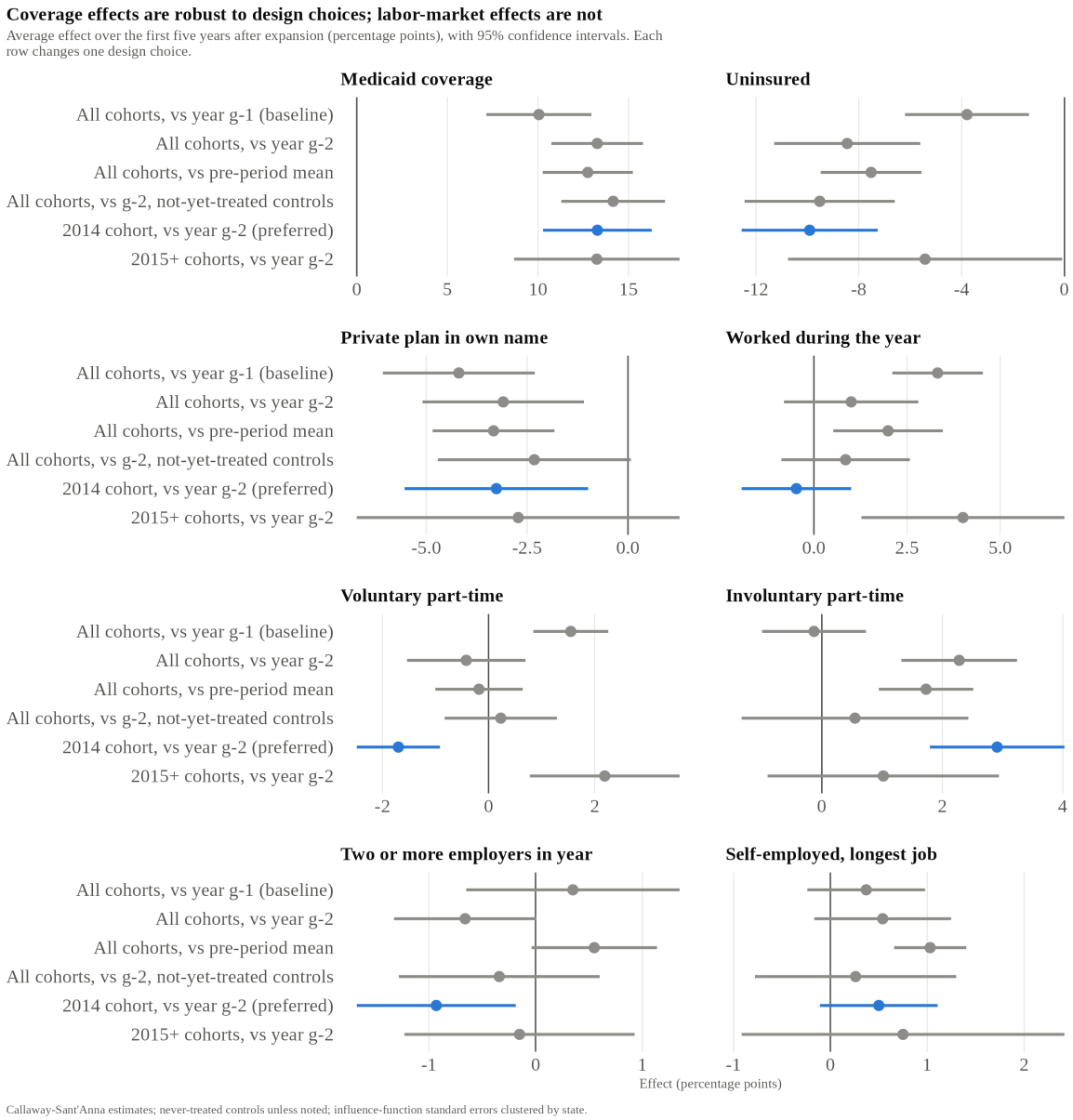


Figure 3. Headline effects under alternative comparison years, control groups, and cohorts

Notes: Average effect over event times 0–4 with 95 percent confidence intervals; values in Table 3. The preferred specification is highlighted.

Table 4. Two-way fixed effects and its decomposition

Panel A. Estimates (average over $e = 0, \ldots, 4$ where applicable)				
	TWFE static	TWFE event study	C–S vs $g-1$	C–S vs $g-2$
Medicaid coverage	13.36	11.67	10.05	13.27
Uninsured	−7.27	−7.73	−3.79	−8.45
Private plan, own name	−3.85	−2.45	−4.19	−3.09
Own employer plan[a]	−1.38	−1.93	−3.83	−1.53
Two or more employers	0.18	0.03	0.35	−0.66
Self-employed	−0.08	0.06	0.37	0.54
Voluntary part-time	−0.02	−0.18	1.55	−0.42
Involuntary part-time	0.14	−0.77	−0.13	2.28
Worked during year	0.89	0.14	3.32	1.00
Annual hours (hours)	15.7	15.0	49.8	8.8
Panel B. Goodman-Bacon decomposition of the static TWFE coefficient (unweighted)				
	Treated vs never-treated	Earlier vs later (clean)	Later vs earlier (forbidden)	Unweighted TWFE
Share of TWFE weight	0.544	0.282	0.175	
Medicaid coverage	12.73	12.84	8.19	11.97
Uninsured	−6.11	−9.32	−6.24	−7.04
Two or more employers	−0.02	0.34	−0.81	−0.05
Self-employed	−0.21	0.21	−0.83	−0.20
Voluntary part-time	−0.06	0.35	0.55	0.16
Worked during year	−0.19	1.28	0.42	0.33

Notes: Panel A: TWFE regressions include state and year fixed effects and use the same state weights as the main estimates; the event-study version pools event times at or below −6 and at or above 8. C–S = Callaway–Sant'Anna, all cohorts, never-treated controls. Panel B: weighted mean of the two-by-two estimates of each type, with each type's share of the TWFE weight in the first row; the decomposition reproduces the unweighted TWFE coefficient exactly. No treated state-year receives a negative TWFE weight.

estimates for those states are not interpreted.

6.3 Samples, timing, and windows

Table 6 and Figure 5 re-estimate the all-cohort specification against year $g-2$ under alternative choices. Coverage results are stable. The Medicaid effect ranges from 11.9 to 15.4 points and the uninsurance effect from −7.1 to −8.7 across the age range, timing definition, sample windows, and weighting; the education-defined sample discussed below is the exception. The labor outcomes are not stable. Annual hours range from +60 hours (sample ending 2019) to −53 hours (ending 2022). The two-or-more-employers estimate ranges from −2.2 to +0.1, and involuntary part-time work from +2.4 to −0.1. No labor outcome shows a statistically significant effect of stable sign across these checks.

One sample deserves comment. Because income-defined samples can change composition if expansion affects earnings, I also define the sample by education (adults without a bachelor's degree, at any income). In that sample, Medicaid coverage rises by 5.5 points, a diluted first stage as expected since most of these adults are above the income limit, and employment rises by 1.5 points (s.e. 0.3). But uninsurance does not fall (−0.3, s.e. 0.7), which is hard to reconcile with a 5.5-point rise in Medicaid and suggests problems in

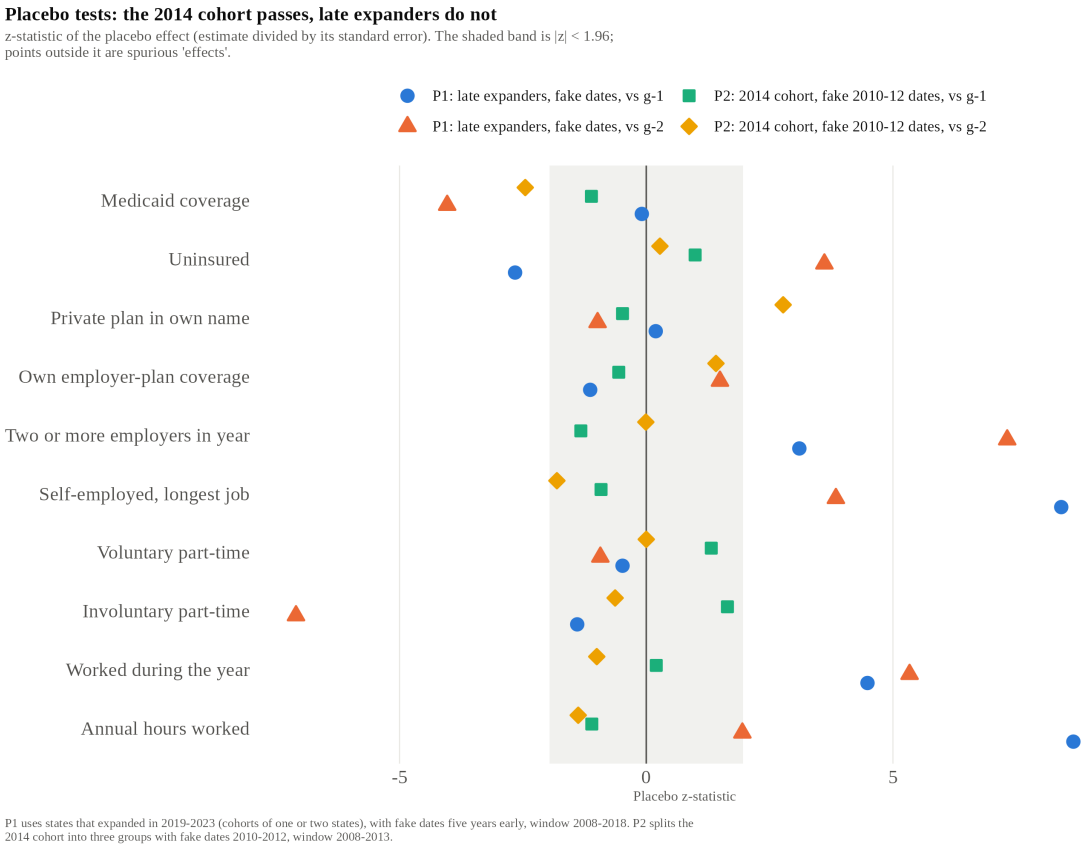


Figure 4. Placebo z-statistics

Notes: Each marker is a placebo estimate divided by its standard error; the shaded band marks $|z| < 1.96$. Values in Table 5.

Table 5. Placebo tests: fake expansion dates

	P2: 2014 cohort, fake 2010–12		P1: late expanders, fake dates	
Outcome	vs $g-1$	vs $g-2$	vs $g-1$	vs $g-2$
Medicaid coverage	-1.99	-4.93^{**}	-0.30	-9.12^{***}
	(1.79)	(2.01)	(3.34)	(2.26)
Uninsured	3.09	1.13	-4.44^{***}	6.03^{***}
	(3.12)	(4.08)	(1.67)	(1.67)
Private plan, own name	-0.52	3.08^{***}	0.39	-1.49
	(1.08)	(1.11)	(2.03)	(1.51)
Own employer plan[a]	-0.58	2.43	-1.83	1.18
	(1.04)	(1.72)	(1.61)	(0.79)
Two or more employers	-1.22	-0.01	2.11^{***}	7.17^{***}
	(0.92)	(1.18)	(0.68)	(0.98)
Self-employed	-0.44	-2.88^{*}	5.72^{***}	3.69^{***}
	(0.48)	(1.59)	(0.68)	(0.96)
Voluntary part-time	1.12	-0.00	-0.77	-0.94
	(0.85)	(0.97)	(1.60)	(1.01)
Involuntary part-time	1.53^{*}	-0.65	-1.33	-3.55^{***}
	(0.93)	(1.03)	(0.95)	(0.50)
Worked during year	0.30	-3.06	8.25^{***}	4.75^{***}
	(1.48)	(3.05)	(1.84)	(0.89)
Annual hours (hours)	-39.9	-42.9	206.5^{***}	40.9^{*}
	(36.2)	(31.1)	(23.9)	(21.0)
Rejections at 5%	0 of 10	2 of 10	5 of 10	6 of 10

Notes: Average placebo effect over all post-placebo event times. P2: the 2014-cohort states are split by state code into three groups with fake dates 2010, 2011, 2012; window 2008–2013; never-treated controls. P1: states that expanded in 2019–2023 receive fake dates five years before their real ones (2014–2018); North Carolina's fake date falls outside the window and it serves as a control; window 2008–2018; real 2014–2016 expanders dropped. With no real treatment, about one rejection in ten is expected by chance at the 5 percent level. [a] Window ends in 2017. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

this sample's comparison. I have not subjected this sample to the full set of pre-trend and placebo checks. I therefore report the estimate without treating it as evidence of an employment effect.

7. Limitations

Several limitations qualify these findings.

Few clusters and small cohorts. Treatment varies at the state level. The preferred specification has 25 states and the all-cohort analysis 40. Standard errors from the influence function are reliable only with many clusters, and they are clearly unreliable for cohorts of one to three states, as the P1 placebo shows. Wild cluster bootstrap or randomization inference would give more credible intervals and is a natural extension.

Remaining pre-treatment differences. Moving the comparison year back removes the level offset, but joint tests still reject zero pre-treatment coefficients for several outcomes, including Medicaid ($p < 0.001$), which reflects one atypical year rather than a trend (slope $p = 0.81$). Self-employment and employer coverage show genuine pre-trends. Formal sensitivity bounds in the spirit of Rambachan and Roth (2023) were

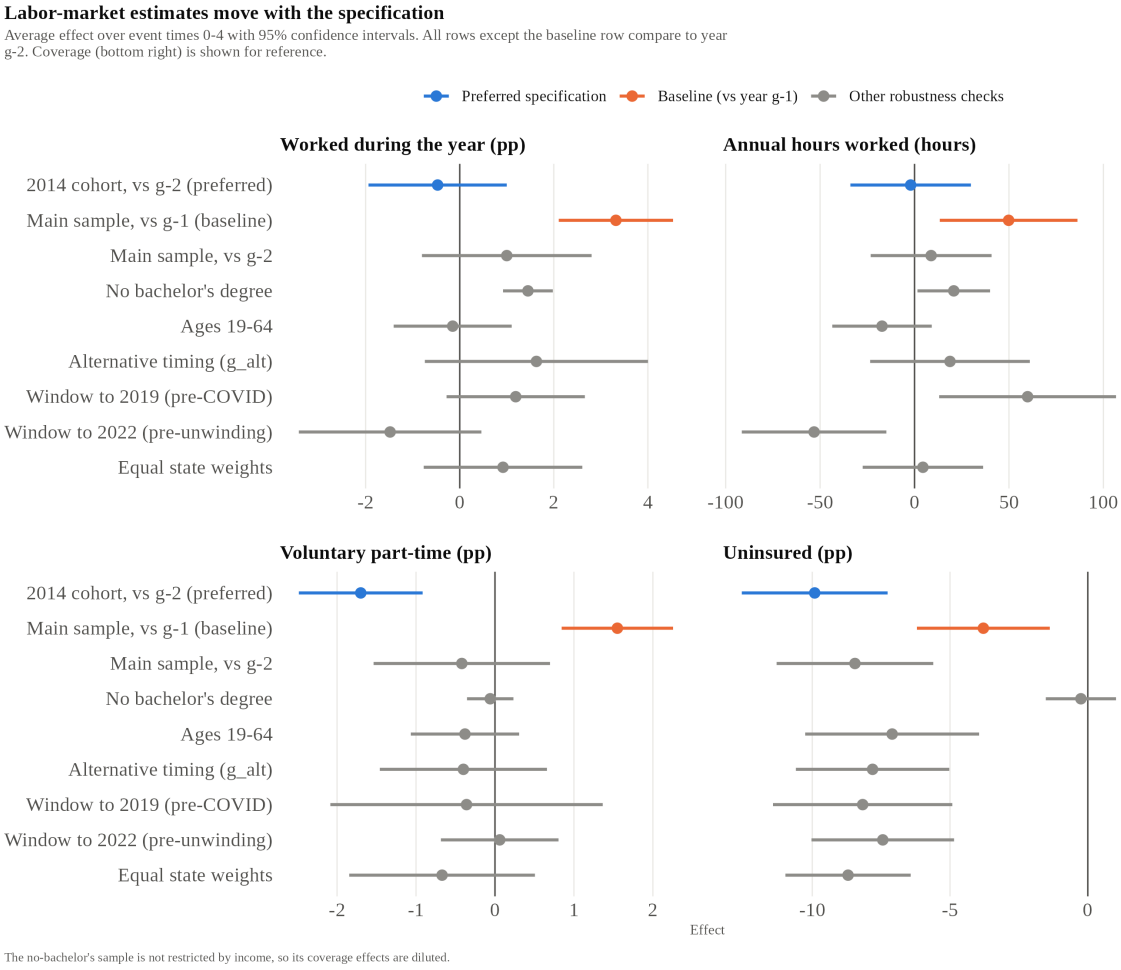


Figure 5. Labor-market and coverage estimates across specifications

Notes: Average effect over event times 0–4 with 95 percent confidence intervals; values in Tables 3 and 6.

Table 6. Alternative samples, timing, and sample windows (all cohorts, comparison year $g-2$)

Outcome	Main sample	No BA degree	Ages 19–64	Alt. timing	Ends 2019	Ends 2022	Equal weights
Medicaid coverage	13.27***	5.54***	11.94***	12.60***	15.11***	15.36***	12.75***
	(1.29)	(0.62)	(1.10)	(1.40)	(1.36)	(1.73)	(1.47)
Uninsured	−8.45***	−0.25	−7.10***	−7.81***	−8.17***	−7.44***	−8.70***
	(1.45)	(0.65)	(1.61)	(1.42)	(1.66)	(1.32)	(1.16)
Private plan, own name	−3.09***	−2.16***	−3.16***	−2.91***	−2.20	−4.24***	−2.28**
	(1.02)	(0.63)	(0.94)	(1.01)	(1.39)	(0.98)	(1.03)
Own employer plan[a]	−1.53*	−0.93	−1.43**	−1.58*	−1.53*	−1.53*	−0.03
	(0.91)	(0.77)	(0.71)	(0.93)	(0.91)	(0.91)	(1.59)
Two or more employers	−0.66*	−0.58***	−0.47	−0.37	−2.22***	0.07	−0.88
	(0.34)	(0.21)	(0.47)	(0.54)	(0.58)	(0.39)	(0.54)
Self-employed	0.54	0.99***	0.57	0.63	−1.60**	0.84***	0.02
	(0.36)	(0.18)	(0.35)	(0.39)	(0.75)	(0.28)	(0.43)
Voluntary part-time	−0.42	−0.06	−0.38	−0.40	−0.36	0.06	−0.67
	(0.57)	(0.15)	(0.35)	(0.54)	(0.88)	(0.38)	(0.60)
Involuntary part-time	2.28***	0.73***	1.57***	2.25***	1.43	−0.08	2.36***
	(0.49)	(0.26)	(0.46)	(0.60)	(0.96)	(0.73)	(0.53)
Worked during year	1.00	1.45***	−0.15	1.63	1.19	−1.48	0.92
	(0.92)	(0.27)	(0.64)	(1.21)	(0.75)	(0.99)	(0.86)
Annual hours (hours)	8.8	20.8**	−17.2	18.8	59.9**	−53.2***	4.4
	(16.3)	(9.8)	(13.4)	(21.6)	(23.9)	(19.5)	(16.3)

Notes: Average effect over $e = 0, \ldots, 4$. "No BA degree" uses adults aged 26–64 without a bachelor's degree at any income. "Alt. timing" assigns each state to the year expansion took effect. "Ends 2019/2022" drops the pandemic and unwinding years; cohorts expanding after the window ends serve as controls. "Equal weights" weights states equally. [a] Reference years 2008–2017 in every column. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

not computed for the preferred specification, and pre-tests have limited power to detect the violations that matter most (Roth, 2022).

Measurement. The CPS-ASEC measures coverage and work retrospectively over a calendar year. Its coverage questions were redesigned in 2014 and its income questions in 2014–2015, at the start of the expansion period. The diagnosis of a partly treated comparison year rests on the estimated year-before coefficients and on the survey's design. It is not directly verified at the individual level. The year-before shifts in part-time work do not have an obvious measurement explanation. Working for two or more employers in a year, top-coded at three, is a crude proxy for job mobility; panel data on job-to-job transitions (the monthly CPS or SIPP) would measure mobility directly.

Sample definition and controls. Defining the sample by income conditions on an outcome that labor supply affects, and the education-defined alternative produced internally inconsistent results. The control states gained subsidized marketplace coverage, which shrinks the measured contrast, and some late-expanding states made their own coverage changes before expanding. The 2020–2023 continuous-enrollment provision and the unwinding affect all states. Ending the sample in 2019 or 2022 leaves the coverage conclusions unchanged (Medicaid 15.1 to 15.4 points, uninsurance −7.4 to −8.2) but moves the labor estimates.

Checks not completed. The code includes a version that drops Wisconsin, which covers adults up to 100 percent of FPL without the expansion, from the control group. In the run reported here that check did not take effect (its estimates were identical to the main sample), so it is not reported. Re-running the HonestDiD

bounds, the education-defined sample's pre-trend and placebo tests, and the Wisconsin exclusion is left for the next revision.

External validity. The preferred estimates describe states that expanded in 2014. These states differ from the late adopters and from the ten states that have not expanded, so effects there could differ.

8. Policy Implications

Coverage. Expansion produced large and durable coverage gains for low-income adults. In the 2014 cohort, about 13 of every 100 low-income adults gained Medicaid and about 10 fewer were uninsured, sustained over five years. Estimates of this size appear in every income-defined specification I examined, except the one that uses the contaminated comparison year.

Work and work requirements. I find no evidence that expansion reduced employment or hours among low-income adults, and the preferred estimate rules out employment reductions larger than about 2 percentage points. This matters for the work requirements that states must implement by January 2027. If expansion did not lower employment, a work requirement is unlikely to raise it much: few people would start working to keep coverage they did not stop working to obtain. That inference is indirect, since this paper estimates the effect of gaining eligibility, not of conditioning it on work. It agrees, however, with direct evidence from Arkansas's 2018 work requirement, which reduced Medicaid coverage with no detectable change in employment (Sommers et al., 2019). It also suggests that the main effect of the new requirement will be on coverage, consistent with CBO's projection of 5.3 million more uninsured people by 2034 (Center for Children and Families, 2025). Agencies designing implementation should weigh compliance and reporting burdens, which drove coverage losses in Arkansas, against small expected employment gains.

Job lock. I find no evidence that expansion released job lock among adults below 138 percent of FPL: job switching, self-employment, and voluntary part-time work did not rise. One reason is that few of these workers held employer coverage in their own name to begin with. Arguments that public coverage frees workers to change jobs or start businesses are more relevant to workers with higher incomes, where employer coverage is common, than to the expansion population.

Measurement practice for analysts. Event studies built on the CPS-ASEC should not use the year immediately before a January policy change as the sole comparison year for coverage outcomes. Comparing with an earlier year, or with the average of the pre-period, and reporting the year-before coefficient explicitly is a simple safeguard. In this application, the default choice would have understated the coverage gain by a quarter to a half and reported a spurious employment increase.

9. Conclusion

The ACA Medicaid expansion substantially increased coverage for low-income adults. It had no robust effect on their employment, hours, job mobility, self-employment, or the voluntary part-time work that job-lock theory predicts. The data rule out meaningful employment declines; the employment estimates range from -0.5 to $+2.0$ points across defensible comparison choices, so a small increase cannot be excluded. Larger estimates, in either direction, depend on a comparison year that is partly treated because of how the CPS-ASEC is fielded, or on cohorts too small to support inference. Modern staggered difference-in-differences methods remove the bias from forbidden comparisons, but they do not protect against a contaminated refer-

ence period; in this application the reference period mattered more than the estimator. The code and data pipeline accompanying this paper reproduce every estimate from the public IPUMS extract.

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Appendix

Table A1. Baseline estimates (comparison year $g-1$) with bootstrap standard errors and pre-trend tests

	Overall effect (simple average)				Pre-treatment tests			
Outcome	ATT	(s.e.)	Treated/control	Window	Wald	p	Slope/yr	p
Medicaid coverage	9.19^{***}	(1.36)	30/10	2008–2024	33.3	< 0.001	0.09	0.718
Uninsured	-3.38^{***}	(1.25)	30/10	2008–2024	9.4	0.093	-0.01	0.959
Private plan, own name	-4.37^{***}	(0.78)	30/10	2008–2024	19.8	0.001	0.13	0.470
Own employer plan[a]	-3.83^{***}	(1.42)	21/19	2008–2017	31.8	< 0.001	-0.43	0.002
Two or more employers	0.63	(0.53)	30/10	2008–2024	21.2	< 0.001	0.18	0.002
Self-employed	0.63^{*}	(0.33)	30/10	2008–2024	51.0	< 0.001	0.25	< 0.001
Voluntary part-time	2.09^{***}	(0.38)	30/10	2008–2024	41.0	< 0.001	-0.35	< 0.001
Involuntary part-time	0.19	(0.46)	30/10	2008–2024	41.6	< 0.001	0.41	< 0.001
Worked during year	2.83^{***}	(0.63)	30/10	2008–2024	11.5	0.042	-0.29	0.022
Annual hours (hours)	27.6^{*}	(16.3)	30/10	2008–2024	5.0	0.415	2.2	0.342

Notes: Callaway–Sant'Anna, all cohorts, never-treated controls, universal base period $g-1$. The overall effect is the cohort-size-weighted average of all post-treatment group-time effects (not restricted to $e \leq 4$), with multiplier-bootstrap standard errors (2,000 draws) clustered by state. Wald: joint test that the five pre-treatment coefficients ($e = -6, \ldots, -2$) are zero (χ^2_5). Slope: generalized least squares slope of the pre-treatment coefficients through $e = -1$. [a] Employer coverage is measured through 2017; expansion cohorts after 2017 serve as controls in that window. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2. TWFE weight by expansion cohort, reference years 2008–2024

Cohort	2014	2015	2016	2019	2020	2021	2022	2023	2024
Share of treated cells	0.640	0.116	0.105	0.047	0.039	0.031	0.012	0.008	0.004
Share of weight	0.564	0.157	0.070	0.071	0.031	0.040	0.037	0.003	0.026

Notes: Implicit weight of each cohort's treated state-years in the weighted static TWFE coefficient. All weights are positive.

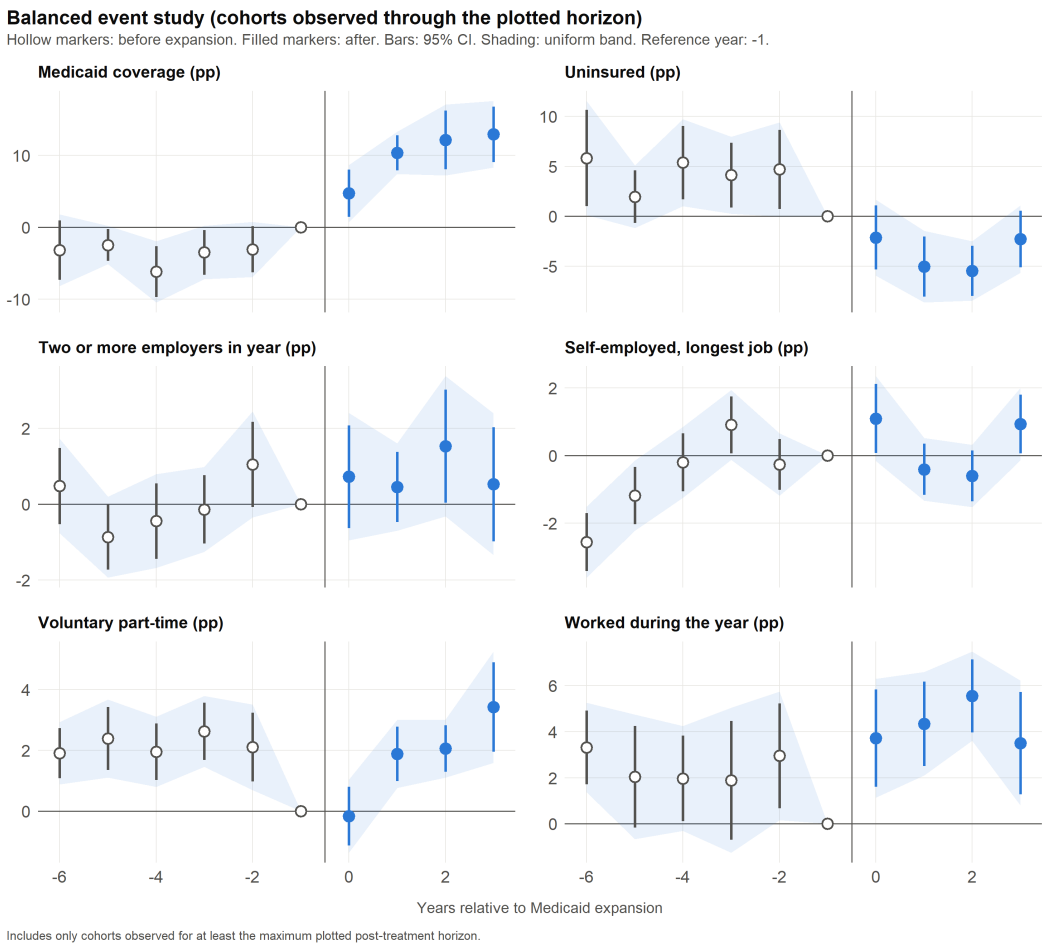


Figure A1. Balanced event study (cohorts observed for at least three post-expansion years), comparison year $g-1$
Notes: Restricting to cohorts observed through $e = 3$ (2014–2021 cohorts) leaves the pre-period offset unchanged, so the offset is not caused by changing cohort composition across event times.

Replication

All estimates are produced by the R scripts in the accompanying repository: 01_clean_data.R (extraction and cleaning), 02_csdid_event_study.R (baseline event studies and pre-trend tests), 03_rebase_and_controls.R (comparison years, control groups, cohorts, placebos), 04_twfe_and_sensitivity.R (TWFE, decomposition, samples and windows), and 05_paper_figures.R (summary figures). Estimation uses the did package (version 2.1.0 or later; results in this paper were produced with version 2.5.1). Results in this paper were produced with 2,000 bootstrap draws for the baseline tables and influence-function standard errors elsewhere.